

Feature 4 (challenge attempts):

Problem: When students generate a study guide, they encounter challenge problems at the end of each topic section. Because the solution is immediately visible, students can read the answer without attempting the problem themselves, creating an illusion of understanding without actual recall or application. This passive consumption pattern undermines the study guide's core value and does not prepare students for exam conditions in which answers are not available.

Target students: STEM students who have uploaded course material and generated a study guide, and who intend to use the challenge problems as practice before a quiz or exam.

Proposed solution: If we let students attempt the challenges at least once before revealing the answer, it will push them to think and apply the concepts. If they fail and look at the answer, they can identify what they did wrong.

Goals:

- **Primary metric:**
 - Quiz scores improve for users who complete challenge attempts, compared with those who skip them. Measured by comparing average quiz scores between the two cohorts in the metric events table.
- **Secondary metric:**
 - Challenge attempt rate, percentage of users who submit at least one attempt per study guide session. The target is 60% of users who view a challenge and submit an attempt.
- **Guardrail metric:**
 - The study guide abandonment rate does not increase after this feature is shipped. If students are leaving the study guide at a higher rate after the challenge attempt gate is introduced, the friction is too high, and the gate needs adjustment.

MVP / Functional Requirements

Core flow:

- [P0] Challenge problems in the study guide are rendered with a text input field and a Submit Attempt button. The solution section is hidden behind a Show Solution button that is disabled until an attempt has been submitted. After submission, both the student's attempt and the full solution are displayed. The Show Solution button becomes active.

- [P0] Attempt persistence: The submitted attempt is stored in session state, so that if the student navigates away and returns, their attempt remains visible and the solution remains unlocked.
- [P1] AI evaluation: After the student submits, an AI-generated response evaluates the attempt. It should identify what the student got right, what they missed, and name the specific concept the gap relates to. This is not a grade; it is formative feedback.
- [P1] Skip option: Students can click "I don't know, show a hint," which counts as a submitted attempt (blank) and unlocks a hint or points to the section in the study guide. This prevents students from being completely blocked if they have no starting point, while still logging that no attempt was made.
- [P2] Attempt quality signal: Track whether the submitted attempt was blank, minimal (under 10 words), or substantive. Use this to refine the guardrail metric and determine whether students are gaming the gate by submitting empty submissions.

Edge case handling:

- Student submits a blank attempt or a single character to bypass the gate, handled by the P2 attempt quality signal. The solution still reveals (we don't block them), but the event is logged as a low-quality attempt, so we can measure how often this happens. If it becomes frequent, it signals the gate needs redesign.
- The student is on mobile with a small keyboard; the text input needs to be large enough to type a meaningful response without the keyboard covering the Submit button.
- The challenge problem is a multiple-choice question rather than an open-ended problem; the text input model doesn't fit. The MVP should only apply to open-ended challenge problems. Multiple-choice challenges should still show the answer immediately since there's no meaningful attempt surface for them.
- The student generates a new study guide for the same material; previous attempts should not carry over, as the guide's content may differ.

Out of Scope:

Grading attempts on a point scale. Storing attempts permanently in the database across sessions. Comparing attempts across users. Generating new challenge problems based on failed attempts (future consideration).